Wong Yen Hong

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Education

- **Multimedia University** — Cyberjaya, Malaysia
  Bachelor of Computer Science (Hons.), Data Science — *Nov. 2022 – 2025*
  - CGPA: 3.93 / 4.00 (First Class Honours)

Publications

- **Yen-Hong Wong***, Lai-Kuan Wong. *AesCrop: Aesthetic-driven Cropping Guided by Composition.* In Proceedings of the Mobile Intelligent Photography & Imaging (MIPI) Workshop at the International Conference on Computer Vision (ICCV), Honolulu, Hawaii, 2025.

Professional Experience

- **Grab** — Petaling Jaya, Malaysia
  Machine Learning Engineer, Pricing — *Jan. 2026 – Present*
  - Developed a city-level network value model using neural networks and temporal-difference (TD) learning to capture expected driver earnings across spatiotemporal states for ride allocation and dynamic pricing; increased order fulfillment rate by +0.5% in offline simulations.
  - Spearheaded a trip-level driver wait-time model using recurrent neural networks (RNNs) with flexible sequence batching for effort-based pricing, boosting prediction accuracy by +12% over the legacy model.
  - Conducted city-level driver earning simulations using a causal inference surge model combined with Bayesian optimization to identify optimal pricing configurations.
  - Extended the in-transit ETA prediction model to support scheduled orders and successfully executed a regional rollout, improving ETA accuracy for scheduled rides.
  - Streamlined ML pipeline operations by expanding unit test coverage, integrating automated monitoring, and optimizing workflow compute efficiency.

  Machine Learning Engineer Intern, AI Platform — *Jul. 2025 – Nov. 2025*
  - Implemented distributed training for the ETA model using Ray Train and Ray Data across a multi-node cluster, achieving a 20× speedup and 85% GPU utilization.
  - Architected a multi-node training framework for metric forecasting via model sharding and activation checkpointing, reducing memory footprint by ∼80%.
  - Built an in-house LLaMA-Factory platform for out-of-the-box LLM/VLM fine-tuning, supporting Ray orchestration, custom S3 artifact management, and internal authentication.
  - Engineered multi-architecture container build pipelines for the internal ML platform to enhance deployment flexibility across diverse target instances.
  - Migrated and deployed production Airflow operators using feature flags to guarantee seamless zero-downtime rollouts.

- **Multimedia University** — Cyberjaya, Malaysia
  Undergraduate Researcher — *Nov. 2024 – Jul. 2025*
  - Conducted research on aesthetic-driven automatic image cropping in the Visual Processing (ViPr) Lab under Associate Prof. Dr. Wong Lai Kuan.
  - Proposed a SOTA hybrid image cropping framework guided by visual composition rules, conducting rigorous experiments to demonstrate aesthetic improvements.
  - Published research findings at the ICCV 2025 MIPI Workshop.

- **MoneyLion** — Kuala Lumpur, Malaysia
  Data Science Intern — *Jul. 2024 – Oct. 2024*
  - Enhanced transaction classification models on long-tail categories to enrich transactional data for downstream financial risk and analytics use cases.
  - Engineered an on-demand rule injection pipeline leveraging clustering, LLMs, and graph traversal to auto-generate deterministic rules, streamlining ground-truth dataset curation.
  - Synthesized 582 deterministic rules that automatically curated ∼14,000 ground-truth labels, increasing categorization accuracy by 30% on long-tail transactions.
  - Optimized Metaflow data pipelines, cutting execution latency 4× (from 8 minutes to 2 minutes) through SQL query tuning and multi-threaded parallel execution.

Projects

- **Counter-Strike Agent**
  - Developed an end-to-end RL agent for Counter-Strike combining offline pre-training on human gameplay with live in-game online fine-tuning.
  - Built custom environment wrappers for real-time state extraction (health, ammo, 2D occupancy grid, target bounding boxes) and action handling (continuous aiming, movement, and firing).
  - Bootstrapped baseline policy via Implicit Q-Learning (IQL) on 4+ hours of curated human demonstrations, using stratified sampling to tackle extreme reward sparsity and accelerate initial learning.
  - Engineered an online fine-tuning loop with dynamic replay buffer experience mixing that smoothly transitions from offline data to stochastic live exploration, accelerating policy convergence and boosting peak episode returns.
  - Evaluated agent decision-making through policy analysis across key behavior axes, demonstrating emergent target alignment, spatial navigation awareness, and obstacle detection.
- **Heuristic Post Ranker**
  - Built a forum post ranking system incorporating net votes, author reputation, comment velocity, and exponential age decay.
  - Designed a constrained Bayesian optimization pipeline using Expected Improvement (EI) and Probability of Feasibility (PoF) to maximize user clicks under recency constraints.
  - Created an LLM-driven multi-persona user behavior simulator with dynamic fatigue mechanisms to evaluate post engagement under realistic traffic conditions.
  - Evaluated the Bayesian-tuned ranker against random, recency-based, and trending baselines, demonstrating superior click-through performance under exhaustion constraints.
- **MakanLah**
  - Curated a 30-class local food image dataset (70 images/class) and developed synthetic adversarial data augmentations for robust classifier training.
  - Trained a MobileNetV2 classifier with Outlier Exposure (OE), achieving 90% in-distribution accuracy and 92% out-of-distribution (OOD) detection capability.
  - Architected a scalable Kubernetes (EKS) MLOps pipeline integrating MLflow (registry), Ray Train/Serve (distributed training and inference), and Metaflow (orchestration).
  - Implemented an automated feedback loop triggering online retraining pipelines based on user telemetry and edge prediction corrections.
- **What If I Never Brick**
  - Developed a competitive programming rating calculator based on ideal contest performance, garnering 400+ upvotes on Codeforces.
  - Implemented a dynamic programming-based optimization strategy alongside the Elo rating algorithm to project performance trajectories.

Conferences & Summer Schools

- **International Conference on Computer Vision (ICCV) 2025** Honolulu, Hawaii
  - Awarded a scholarship via the Broadening Participation Program to attend and present research as an undergraduate author.
- **Oxford Machine Learning Summer School (OxML) 2025** Oxford, United Kingdom
  - Selected for student scholarship in the *MLx Representation Learning & Generative AI* Track at the University of Oxford's Mathematical Institute.

Honors and Awards

- **Codeforces**: Ranked top 4% of users on the world's largest competitive programming platform.
- **Meta Hacker Cup 2023**: Secured top 4% in Round 1 (824 / $\approx$ 20,000).
- **Final Year Project**: Received the Best Project Award in Data Science (1 / $\approx$ 230).
- **Merit Scholarship**: Awarded with a merit scholarship for outstanding foundation result.
- **Programming League National 2025**: Secured 2nd place among 100+ teams.

- **Programming League National 2024**: Secured 1st place among 70+ teams.
- **Programming League National 2023**: Earned 3rd place among 60+ teams.
- **Monash Coding League 2024**: Secured 1st place among 80+ teams.
- **Monash Coding League 2023**: Won 1st place among 70+ teams.
- **CodeNection 2022**: Achieved 2nd place among 100+ teams.
- **CodeNection 2021**: Achieved 1st place in closed category among 30+ teams.
- **ImagineHack 2024**: Earned Best Innovation Award.
- **Data Science Digital Race 2024**: Won 1st place among 100+ teams.

Technical Skills

- **Languages & Frameworks**: Python (PyTorch, TensorFlow, Ray, Metaflow, Pandas, FastAPI, OpenCV, Spark), JavaScript/TypeScript (React.js, Next.js, Node.js, Express.js), C++, Java, SQL
- **MLOps, Cloud & Tools**: AWS, Terraform, Kubernetes, Docker, MLflow, Ray Serve, Git, GitHub Actions, Linux

Volunteering & Leadership

- **CodeNection (2023 – 2024)**: Competition Lead & Problem Author; authored 20+ algorithmic problems and led team of 50 for 700+ participant competition.
- **Programming League National 2025**: Invited Workshop Speaker on Advanced Algorithms for 150+ participants (4.7/5 rating).
- **Hackerspace MMU**: Active Member.